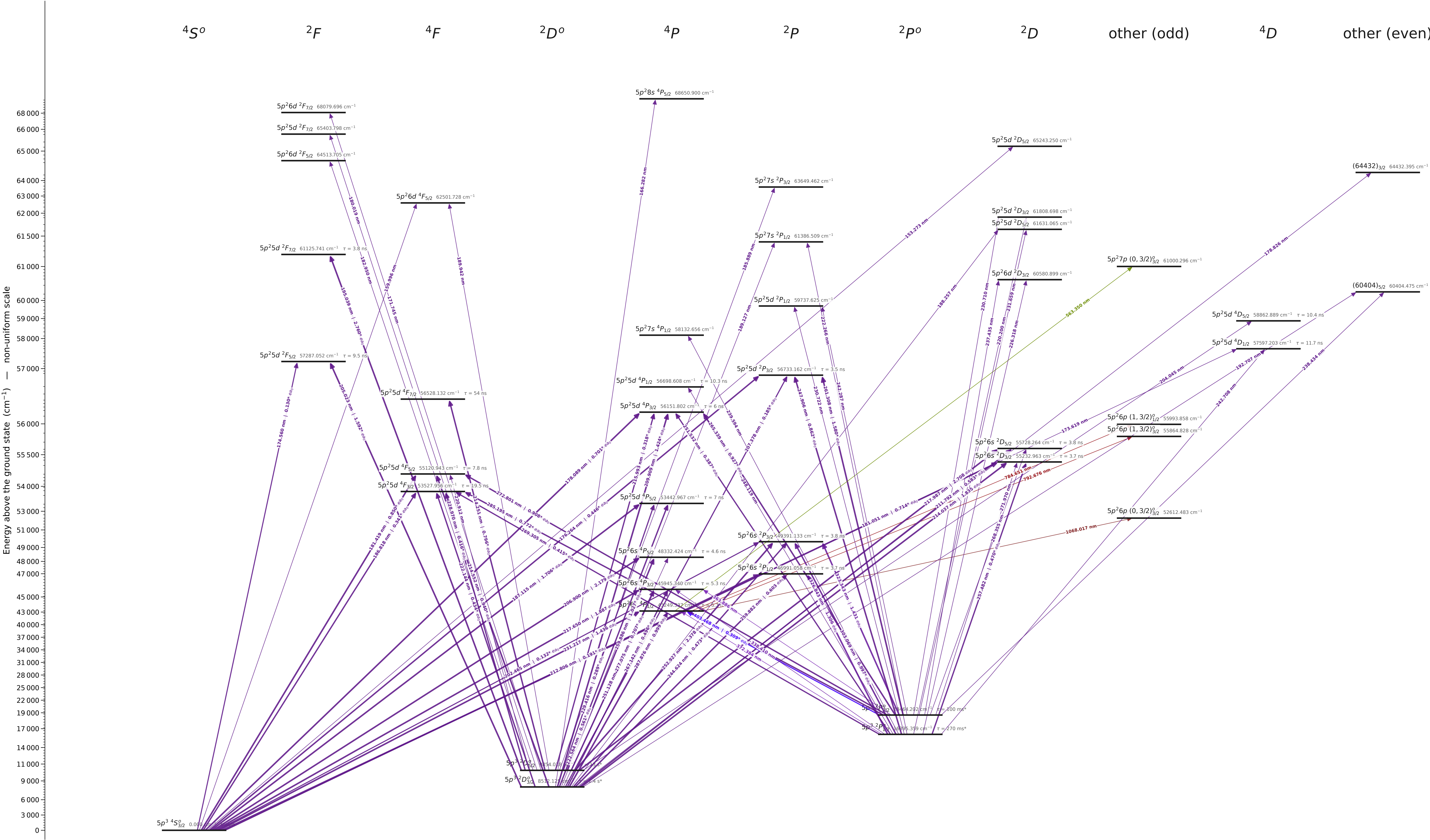
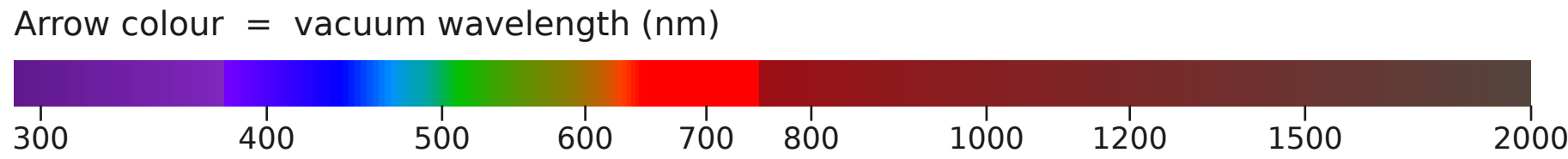


Antimony

Sb (Sb I). Strongest classified transitions below 2 μm (NIST): 42 levels, 82 lines.

Sb ⁺ ionisation limit 69431.34 cm⁻¹ = 8.60839 eV (144.027 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||e r||U'|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U e r U' $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5p ³ ⁴ S _{3/2} - 5p ² 6s ⁴ P _{3/2}	206.9004	206.8344	1448.9696	2.179	1.810e+08	34.6	90.4	laser pumping line
5p ³ ⁴ S _{3/2} - 5p ² 6s ⁴ P _{3/2}	217.6499	217.5818	1377.4066	1.887	1.750e+08	30.03	88.9	ground-state line used for collinear laser spectroscopy o...
5p ³ ⁴ S _{3/2} - 5p ² 6s ⁴ P _{1/2}	231.2174	231.1463	1296.5825	1.436	1.690e+08	30.03	82.5	laser pumping line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹²¹Sb hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
5p ³ ⁴ S _{3/2}	-299.034(4)	-3.680(20)	F=1: +1567.35 F=2: +972.23 F=3: +76.78 F=4: -1122.30
5p ³ ² D _{3/2} ^o	565(7)	-561(6)	F=1: -3358.95 F=2: -1780.15 F=3: +167.30 F=4: +1978.50
5p ³ ² D _{5/2}	1468.980	-	F=0: -12853.58 F=1: -11384.59 F=2: -8446.64 F=3: -4039.70 F=4: +1836.22 F=5: +9181.12
5p ³ ² P _{1/2} ^o	4907.600	-	F=2: -8588.30 F=3: +6134.50
5p ³ ² P _{3/2}	680.0(10)	438(30)	F=1: -3263.40 F=2: -2253.80 F=3: -410.90 F=4: +2659.50
5p ² 6s ⁴ P _{3/2}	523.8(5)	-471.0(20)	F=1: -3079.65 F=2: -1655.25 F=3: +128.10 F=4: +1846.50

Measured A, B constants; sources in the data file.